# Results outline corintreespotters

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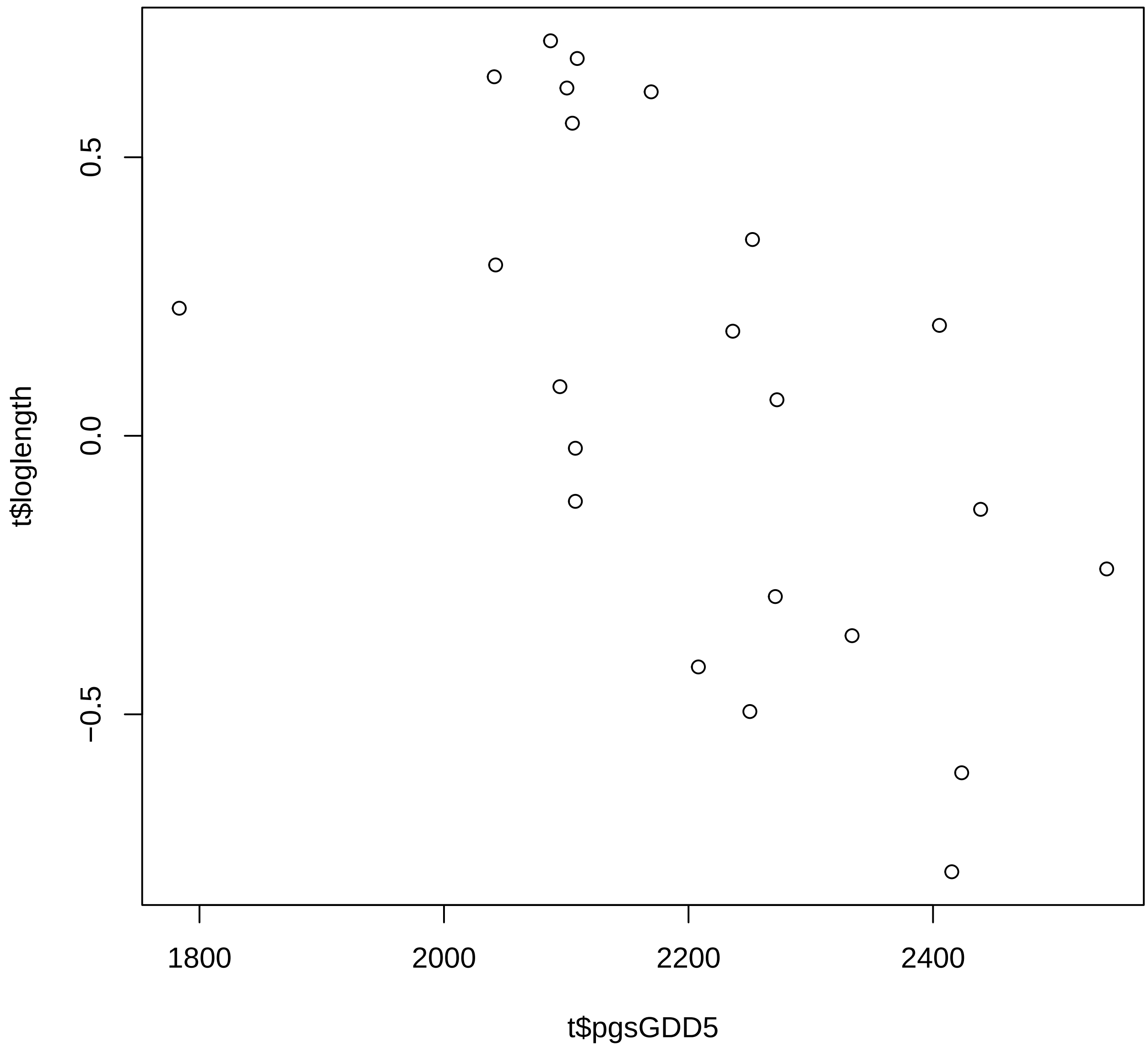


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## Error in `h()`:
## !  error in evaluating the argument 'y' in selecting a method for function 'plot':  object 'y_pos'
not found
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## Research questions

1. Does tree growth increase with longer growing seasons?
2. Did an earlier start and later end of season increase growth?
3. Does tree growth increase with longer growing seasons and is it consistent across species?
4. How did growth change across years, and how did it relate to climate?

## Results

1. CoringTreespotters includes eleven deciduous tree species with phenological observations from 2016 to 2024. In total, our final dataset has 329 leafout observations and 310 leaf coloring observations. We used these phenophases to set the boundaries of individual-level growing seasons, which together yield 280 complete growing seasons.

1. Summary basic information
   (a) Coringtreespotters includes eleven with leaf phenology data spaning nine years (2016 to 2024) for a total of 329 leafout and 310 colored leaves observations, yielding 280 complete growing seasons (GS) for 49 individuals.
   (b) GSL ranged from 77 days (*A. flava* in 2018) to 194 days (*A. saccharum* in 2021)
   (c) GS thermal sum ranged from 1048.8 GDD (*A. flava* in 2020) to 2634.7 GDD, (*T. americana* in 2016)
   (d) The earliest leafout day of year was 08-Apr (2021) and the latest, 14-Jun (2021). The earliest coloredLeaves day of year was 15-Jul (2021) and the latest, 06-Nov (2021)
   (e) 2018 had a mean summer temperature of 22.85°C and had a mean summer PDSI value of 1.92. 2019 was cooler (22.38°C) and wetter (PDSI = 4.02), whereas 2020 was warmer (23.07°C) and drier (PDSI = -2.35) than 2018.

2. Does tree growth increase with longer growing seasons?
   (a) No clear trend of growing season length effect across our eleven species and season predictors.
   (b) Longer thermal season increased growth *weakly for *A. saccharum*($\beta$ = 0.09, $UI_{90}$ [-0.02, 0.2]), *B. nigra*($\beta$ = 0.08, $UI_{90}$ [-0.01, 0.16]), *C. glabra*($\beta$ = 0.05, $UI_{90}$ [-0.06, 0.17]), *Q. rubra*($\beta$ = 0.08, $UI_{90}$ [-0.09, 0.26]), and decreased for *B. alleghaniensis* ($\beta$ = -0.22, $UI_{90}$ [-0.36, -0.09]) ) (Figure **??**a and e) (Table **??**).
   (c) Of the four species that grew more with longer thermal season all, only two also grew more with longer calendar season (*B. nigra*: $\beta$ = 0.03, $UI_{90}$ [-0.02, 0.07], *C. glabra*: $\beta$ = 0.03, $UI_{90}$ [-0.03, 0.08]), and *T. americana* growth decreased ($\beta$ = -0.02, $UI_{90}$ [-0.05, 0.01]), unlike its response to the thermal season.
   (d) *B. alleghaniensis*trended in the same direction, but showed a weaker reponse than the thermal season ($\beta$ = -0.05, $UI_{90}$ [-0.11, 0.02]) (Figure **??**b and f) (Table **??**). (Figure **??**b).
   (e) Results with basal area increment (BAI) for the GDD model were broadly similar to ring width (see Figure **??** and Table **??**)

3. Did a later start and end of season increase growth?
   (a) A later start of season increased growth for *T. americana* ($\beta$ = 0.14, $UI_{90}$ [0.01, 0.26]), but *A. saccharum*: $\beta$ = 0.07, $UI_{90}$ [-0.02, 0.16], *B. nigra*: $\beta$ = 0.03, $UI_{90}$ [-0.1, 0.16], *C. glabra*: $\beta$ = 0.07, $UI_{90}$ [-0.09, 0.22], *C. ovata*: $\beta$ = 0.09, $UI_{90}$ [-0.04, 0.22], *P. deltoides*: $\beta$ = 0.06, $UI_{90}$ [-0.08, 0.2], *Q. rubra*: $\beta$ = 0.08, $UI_{90}$ [-0.09, 0.26] trended in the same direction, and *B. alleghaniensis*($\beta$ = -0.07, $UI_{90}$ [-0.22, 0.07]) trended in the opposite direction (Figure **??**c and g) (Table **??**).
   (b) A later end of season decreased growth for *B. alleghaniensis* ($\beta$ = -0.08, $UI_{90}$[-0.15, -0.01]), but *T. americana*($\beta$ = -0.01, $UI_{90}$[-0.04, 0.02]) trended towards the same direction and *A. saccharum* ($\beta$ = 0.03, $UI_{90}$ [-0.03, 0.08], *B. nigra* ($\beta$ = 0.03, $UI_{90}$[-0.02, 0.08]), and *C. glabra*: $\beta$ = 0.03, $UI_{90}$ [-0.02, 0.09] trended towards an increase in growth. (Figure **??**d and h) (Table **??**).

../../analyses/figures/growthModelsMain/TSmuALLbsppWlines.jpeg

Figure 1: Model slope estimates representing ring width change as a function of four different predictors across four species. GDD represent the accumulated growing degree days between leafout and coloredLeaves and the models estimates shows ring width change per 7 average spring days GDD (176.87) (a). GSL is the number of days between leafout and coloredLeaves (b). SOS represent the start of season, which is leafout (c) and EOS, the end of season through coloredLeaves (d). All predictors were fit using the same model and number of observations, with only the independent variable changing. The dots represent the mean of each species and the lines, the 50% and 90% uncertainty intervals (UI).

../../analyses/figures/growthModelsMain/TSmuayear.jpeg

Figure 2: We show tree ring widths (mm) across years as model estimates for the year intercepts with the dots depicting the mean and the lines, the 50% and 90% uncertainty intervals (UI).

## Supplemental results

0S0 0S0

../../analyses/figures/growthModelsMain/TSgrowthModelSlopesperSppFacetGDD.jpeg

Figure 1: Ring width trends in response to thermal season (GDD). The plots are the reconstructed species estimates posterior distribution with error. The lines represent the mean of the posterior distribution and the shaded area, the 50% UI. The dots are the empirical data shaped by year.

../../analyses/figures/growthModelsMain/TSgrowthModelSlopesperSppFacetGSL.jpeg

Figure 2: Ring width trends in response to calendar season (GSL). The plots are the reconstructed species estimates posterior distribution with error. The lines represent the mean of the posterior distribution and the shaded area, the 50% UI. The dots are the empirical data shaped by year.

../../analyses/figures/growthModelsMain/TSmeanPlotGrowthGDD_treeidBYspp.pdf

Figure 3: Intercept estimates for each group of the GDD model. The different tree id values represent the sum of the tree id, provenance, species and averaged year intercepts. The larger dots and lines represent aggregated average across these values. The dots are the mean of the posterior distribution and the thick and thin lines, the 50 and 90% UI.